



Day 24 — ReAct Prompting & Structured Prompting



Introduction

In previous lectures, we learned:

- Prompt Engineering
- Zero-Shot / Few-Shot Learning
- Chain-of-Thought (CoT)

Now the next evolution is:

👉 ReAct Prompting

Modern AI systems do not just:

- generate text
- answer questions

They can now:

- reason
- use tools
- interact with systems
- retrieve information
- perform actions

This capability is powered by:



ReAct Prompting

Along with this, modern production AI systems also require:

👉 Structured Prompting

because enterprises need:

- reliable outputs
- predictable formatting
- automation-friendly responses

◆ 1. What is ReAct Prompting?



Definition

ReAct stands for:

👉 Reason + Act

It is a prompting technique where the model:

1. reasons about the problem
2. decides an action
3. performs the action using a tool
4. observes the result
5. reasons again
6. generates final answer

Core Idea

Traditional LLM:

Plain text



Only thinks.

ReAct model:

Plain text



Thinks + interacts with environment.

◆ 2. Why ReAct Became Important

Earlier LLMs:

- only generated text

But modern AI systems require:

- web search
- API calls
- calculators
- databases
- code execution

LLMs alone cannot reliably solve these tasks without tools.

So ReAct was introduced to combine:

 **reasoning + actions**

◆ 3. ReAct Workflow

🧠 Standard ReAct Flow

Plain text



Thought → Action → Observation → Thought → Final Answer

🔍 Step-by-Step Meaning

Step	Meaning
Thought	Analyze problem
Action	Use tool/API
Observation	Read result
Final Answer	Generate response

◆ 4. Simple ReAct Example

? User Question

Plain text

What is Tesla stock price today?

🧠 Model Reasoning

Plain text

Thought:
I need latest stock data.

⚡ Action

Plain text

Action:
Use stock market API/web search.

👁️ Observation

Plain text

Observation:
Tesla stock price = \$XXX

◆ 5. Difference Between CoT and ReAct

Chain-of-Thought (CoT)

Only reasoning

Internal thinking

Static answers

No environment interaction

ReAct

Reasoning + actions

External tool interaction

Dynamic workflows

Uses APIs/tools

🔥 Important Understanding

🧠 CoT

= Think step-by-step

🧠 ReAct

= Think + Take Action

◆ 6. Why ReAct is Powerful

ReAct allows models to:

- retrieve latest data
- perform calculations
- search documents
- use APIs
- execute code
- interact with external systems

◆ 7. Tools Used in ReAct Systems

Modern AI systems can use tools like:

Tool	Purpose
Calculator	Math operations
Web Search	Latest information
Python	Code execution
Database	Retrieve data
APIs	External systems
Browser Tools	Website interaction

◆ 8. Real-Life ReAct Example



Travel Assistant

User asks:

Plain text

Find cheapest flight from Delhi to Dubai tomorrow.



AI Reasoning

Plain text

Thought:
Need live flight information.

Action

Plain text

Use flight search API.

Observation

Plain text

Received flight data.

Reason Again

Plain text

Compare prices and timings.

Final Answer

Plain text

Cheapest flight is XYZ Airlines at ₹18,000.

◆ 9. ReAct in Modern AI Systems (2026)

Modern reasoning models like:

- GPT-5 systems
- Claude reasoning models
- Gemini agentic systems

use:

- reasoning
- planning
- tool orchestration
- multi-step workflows

ReAct principles are foundational to these systems.

◆ 10. What is Structured Prompting?

✓ Definition

Structured Prompting means:

Designing prompts in an organized and controlled way to get predictable outputs.

🔍 Core Idea

Instead of vague instructions:

- use structured templates
- define output format
- specify constraints

◆ 11. Why Structured Prompting is Important

In production AI systems:
random outputs are dangerous.

Applications need:

- consistency
- automation
- reliability
- machine-readable outputs

◆ 12. Unstructured vs Structured Prompting

✗ Unstructured Prompt

Plain text

Tell me about Apple company.

Output may vary every time.

✓ Structured Prompt

Plain text

Summarize Apple company in JSON format with:

- founder
- CEO
- headquarters
- products

Now output becomes:

- predictable
- structured
- automation-friendly

◆ 13. Components of Structured Prompting

Component	Purpose
Role	Define AI identity
Instruction	Define task
Context	Background info
Constraints	Rules/limits
Output Format	Expected structure

◆ 14. Role Prompting

✅ Definition

Assigning a role/persona to the model.

🔍 Example

Plain text

You are a senior financial analyst.



🧠 Why Useful?

Helps model:

- adapt tone
- use domain knowledge
- improve context understanding

◆ 15. Output Formatting

Modern AI systems often require outputs in:

- bullet points
- tables
- markdown
- JSON
- XML

🔍 Example

Plain text

Return answer in JSON format only.

◆ 16. Prompt Constraints

Constraints improve reliability.

🔍 Examples

Plain text

Answer in 3 bullet points only.

Plain text

Use beginner-friendly English.

Plain text

Do not exceed 100 words.

◆ 17. Why Structured Prompting Matters in APIs

Production systems require:

- parsable outputs
- reliable formatting
- consistent responses

Without structure:
automation breaks.

◆ 18. JSON Prompting

VERY IMPORTANT modern topic.

🔍 Example

Plain text

```
Extract user details in JSON format:
{
  "name": "",
  "email": "",
  "phone": ""
}
```

🧠 Why Important?

Used in:

- APIs
- workflows
- automation systems
- enterprise AI pipelines

◆ 19. Common Problems Without Structured Prompting

Without structure:

- inconsistent output
- hallucinated fields
- broken JSON
- automation failure

◆ 20. Structured Prompting Best Practices

✅ Be Specific

Bad:

Plain text

Explain AI.

Good:

Plain text

Explain transformers for beginners in 5 bullet points.

✅ Define Output Format

Specify:

- JSON
- markdown
- table
- bullets

◆ 21. Real-World Applications

Customer Support Systems

Structured ticket generation.

Financial AI

Structured report analysis.

Medical AI

Standardized patient summaries.

AI Automation Systems

Machine-readable outputs.

◆ 22. ReAct + Structured Prompting Together

Modern AI systems combine:

- reasoning
- tool usage
- structured outputs

Example:

1. reason
2. use tool
3. retrieve data
4. return structured JSON

This is core architecture of modern AI products.
